1 I

1.1 (1)

$$L_1 = T - V = \frac{1}{2}m_1\dot{y}_1^2 + \frac{1}{2}m_2\dot{y}_2^2 + m_1gy_1 + m_2gy_2.$$

1.2 (2)

$$y_1 + y_2 + \pi a = l,$$
$$C_1(y_1, y_2) = y_1 + y_2 + \pi a - l = 0.$$

1.3 (3)

$$L_1' = L_1 + \lambda_1 C_1$$
$$= \frac{1}{2}m_1\dot{y}_1^2 + \frac{1}{2}m_2\dot{y}_2^2 + m_1gy_1 + m_2gy_2 + \lambda_1(y_1 + y_2 + \pi a - l).$$

Applying the Euler–Lagrange equations to y_1 and y_2, respectively, gives

$$m_1\ddot{y}_1 = m_1g + \lambda_1,$$
$$m_2\ddot{y}_2 = m_2g + \lambda_1.$$

1.4 (4)

$$\ddot{y}_1 = \frac{m_1 - m_2}{m_1 + m_2}g.$$

1.5 (5)

$$\lambda_1 = -\frac{2m_1m_2}{m_1 + m_2}g.$$

With the chosen sign convention for C_1, the rope tension is $T = -\lambda_1$.

2 II

2.1 (6)

$$I = \int r^2\,dm = \int_0^a \sigma r^2(2\pi r\,dr) = \frac{1}{2}Ma^2.$$

2.2 (7)

$$T_3 = \frac{1}{2}I\dot{\theta}^2 = \frac{1}{4}Ma^2\dot{\theta}^2.$$

2.3 (8)

$$y_1 = a\theta,$$
$$C_2(y_1, \theta) = y_1 - a\theta = 0.$$

2.4 (9)

$$L_2 = \frac{1}{2}m_1\dot{y}_1^2 + \frac{1}{2}m_2\dot{y}_2^2 + \frac{1}{4}Ma^2\dot{\theta}^2 + m_1gy_1 + m_2gy_2.$$

$$L_2' = L_2 + \lambda_1(y_1 + y_2 + \pi a - l) + \lambda_2(y_1 - a\theta).$$

Applying the Euler–Lagrange equations to y_1, y_2, and θ, respectively, gives

$$m_1\ddot{y}_1 = m_1g + \lambda_1 + \lambda_2,$$
$$m_2\ddot{y}_2 = m_2g + \lambda_1,$$
$$I\ddot{\theta} = -a\lambda_2.$$

2.5 (10)

$$\ddot{y}_1 = \frac{m_1 - m_2}{m_1 + m_2 + M/2}g.$$

2.6 (11)

$$\lambda_1 = -\frac{m_2(2m_1 + M/2)}{m_1 + m_2 + M/2}g,$$
$$\lambda_2 = -\frac{M}{2}\frac{m_1 - m_2}{m_1 + m_2 + M/2}g.$$

Let T_1 and T_2 denote the tensions on the y_1 and y_2 sides of the rope, respectively. With the chosen sign convention, $\lambda_1 = -T_2$ and $\lambda_2 = T_2 - T_1$. Therefore, λ_1 represents the negative of the rope tension on the y_2 side, while λ_2 represents the difference between the tensions on the two sides of the pulley, which is responsible for its rotational motion.